

Vapnik-Chervonenkis Dimension, Probabilistic Bounds and Application to Neyman-Pearson Classification

Jiajun Zhang and Emile Petit
MATH 598 Presentation

October 27, 2025

Type I & II Error

Let $\mathcal{D}_n := \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ be the training set, $(\mathbf{x}_i, y_i)$ are jointly distributed random variables from the probability space $\mathcal{X} \times \mathcal{Y}$. We consider a binary classification problem where $y_i \in \{0, 1\}$. A **classifier** is a Borel-measurable function $f : \mathcal{X} \rightarrow \mathcal{Y}$. Our goal is to find the “best” classifier based on the data we have and make predictions on other data.

- Misclassification occurs when $f(\mathbf{x}_i) \neq y_i$, it can be summarized into two types of misclassification:
 - **Type I Error:** $\mathbb{P}(f(\mathbf{x}_i) = 1 | y_i = 0)$, namely, the conditional probability that the predicted label is 1 given that the true label is 0.
 - **Type II Error:** $\mathbb{P}(f(\mathbf{x}_i) = 0 | y_i = 1)$, namely, the conditional probability that the predicted label is 0 given that the true label is 1.
- We use $R_0(f), R_1(f)$ to denote the type I, II error made by classifier f , respectively.

Minimizing the Risk

In order to find the “best” classifier, the classical binary classification aims to minimize the weighted sum of type I,II error:

$$f^* := \arg \min_f \mathbb{P}(y = 0)R_0(f) + \mathbb{P}(y = 1)R_1(f)$$

where we denote $\pi_0 := \mathbb{P}(y = 0)$, $\pi_1 := \mathbb{P}(y = 1)$ be some priori probabilities of the class 0, 1.

However, this minimization is not always optimal.

Minimizing the Risk

- Consider an application in cancer diagnosis. If we let $y = 0$ be the class of healthy people, and $y = 1$ be the class of diseased people. Then the consequences of type I & II error may be different, i.e. misclassify a diseased person as healthy may be more serious than misclassify a healthy person as diseased!
- Hence in some cases we are more interested in controlling one of $R_0(f)$, $R_1(f)$ below a certain level.
- In Neyman-Pearson classification, we will manually set up a threshold α , such that we aim to minimize $R_1(f)$ under the constrain that $R_0(f) \leq \alpha$ or vise versa. This is the **NP Oracle Classifier**:

$$f^* = \operatorname{argmin}_{f, R_0(f) \leq \alpha} R_1(f)$$

or

$$f^* = \operatorname{argmin}_{f, R_1(f) \leq \alpha} R_0(f)$$

Neyman-Pearson Lemma

- The NP Oracle Classifier f^* can be obtained by Neyman-Pearson lemma:

$$f^*(\mathbf{x}) := \begin{cases} 1 & \text{if } \frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} > \eta \\ 0 & \text{otherwise} \end{cases}$$

where $f_0(\mathbf{x}), f_1(\mathbf{x})$ are the density from two classes and η satisfies

$$\mathbb{P} \left(\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} > \eta \middle| y = 0 \right) = \alpha$$

- So if the density of $f_0(\mathbf{x}), f_1(\mathbf{x})$ are given, then NP-lemma will give us the optimal classifier for a given significance level α .
- We are more interested in the case when their density is not known, i.e we only know $f_j, j \in \{0, 1\}$ by a training set $\mathcal{D}_n$.
- Two classical techniques are **Empirical Risk Minimization Approach** and **Plug-in Approach**.

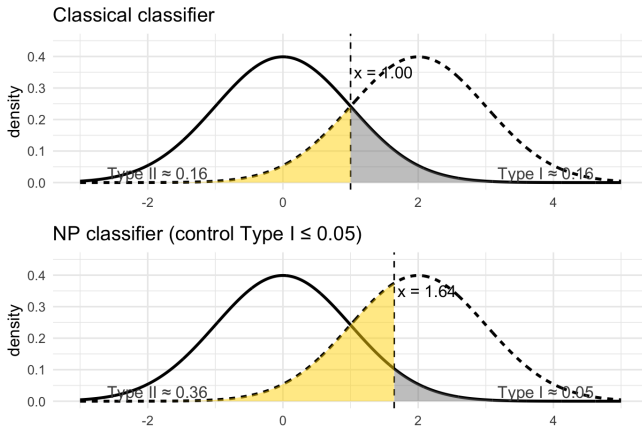


Figure: A simulation: The true distributions of $\mathbf{x}$ under two classes are $\mathcal{N}(0, 1)$ (with $y = 0$), $\mathcal{N}(2, 1)$ (with $y = 1$). If user prefers a type I error $R_0(f) \leq 0.05$, a classical classifier $\mathbf{1}\{x \geq 1\}$ would have $R_0(f) = 0.16$, but the NP classifier $\mathbf{1}\{x \geq 1.65\}$ will satisfy the type I error constraint while minimize type II error.

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Let $f : \mathcal{X} \rightarrow \{0, 1\}$ be a binary classifier, let $S \subset \mathcal{X}$, we define

$$f|_S : S \rightarrow \{0, 1\}, \quad f|_S(\mathbf{x}) = f(\mathbf{x}), \forall \mathbf{x} \in S$$

as the restriction of f to S . Let $\mathcal{C}$ be the class of all classifiers, we define

$$\Pi_{\mathcal{C}}(S) := \left\{ f|_S : f \in \mathcal{C} \right\}$$

as the number of distinct restrictions of classifiers in $\mathcal{C}$ defined by S , i.e., the set of all possible dichotomies on S :

$$\Pi_{\mathcal{C}}(S) := \{(f(\mathbf{x}_1), \dots, f(\mathbf{x}_m)), f \in \mathcal{C}\}$$

where we assume $S := \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ and $f(\mathbf{x}_i) \in \{0, 1\}$. Clearly $|\Pi_{\mathcal{C}}(S)| \leq 2^{|S|}$.

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Definition: A finite set $S \subset \mathcal{X}$ is *shattered* by $\mathcal{C}$, if $|\Pi_{\mathcal{C}}(S)| = 2^{|S|}$. The *Vapnik-Chervonenkis Dimension* (VCD) of $\mathcal{C}$, denote as $\mathbf{VCD}(\mathcal{C})$ (or simply V) is the cardinality d of the largest set S shattered by $\mathcal{C}$. We say $\mathbf{VCD}(\mathcal{C}) = \infty$ if $\mathcal{C}$ shatters arbitrarily sets.

- In other words, $S = \{\mathbf{x}_1, \dots, \mathbf{x}_m\} \subset \mathcal{X}$ is shattered by $\mathcal{C}$ if for every assignment of labels to those points, there exists $f \in \mathcal{C}$ such that it makes no errors when evaluating that set of data points.

We also define the **Growth Function** as

$$\Pi_{\mathcal{C}}(m) = \max\{|\Pi_{\mathcal{C}}(S)|, |S| = m\}$$

where $m \in \mathbb{N}$. If $m \leq \mathbf{VCD}(\mathcal{C})$, we simply have $\Pi_{\mathcal{C}}(m) = 2^m$.

Some Examples:

- **Heaviside Step Functions:** Consider the class of classifiers $\mathcal{C}$ defined by

$$H_{\theta}(x) = \begin{cases} 0 & \text{if } x < \theta \\ 1 & \text{otherwise} \end{cases}$$

for $\theta \in \mathbb{R}$, then we claim that $\mathbf{VCD}(\mathcal{C}) = 1$.

The idea is that, a threshold classifier (like Heaviside step function) can shatter any single point; however it cannot shatter all sets with two points. Consider $S = \{x_1, x_2\}$ and wlog let $x_1 < x_2$. Then if we assign label 1 to x_1 then all classifiers f must assign label 1 to x_2 as well so no such errorless f exists for the label $x_1 = 1, x_2 = 0$.

Some Examples:

- **Interval Classifiers:** Consider the class of classifiers $f \in \mathcal{C}$ defined by

$$f(x) := \begin{cases} 1 & \text{if } x \in [\alpha, \beta] \\ 0 & \text{otherwise} \end{cases}$$

for $\alpha, \beta \in \mathbb{R}$, we then claim that $\mathbf{VCD}(\mathcal{C}) = 2$, i.e. an interval classifier can shatter any sets of two points.

- It is not possible for it to shatter all sets of three points, because if $x_1 < x_2 < x_3$ and x_1, x_3 has label 1 then x_2 must also have label 1 so no such classifier exists for the label $x_2 = 0$ in this case.

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Some Examples:

- **Rectangles in $\mathbb{R}^2$:** Consider $\mathbf{x} \in \mathbb{R}^2$ and define $\mathcal{C}$ as the class of all rectangles:

$$f(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{x} \in [\alpha_1, \beta_1] \times [\alpha_2, \beta_2] \\ 0 & \text{otherwise} \end{cases}$$

We may assume any three points are not co-linear because if so then it reduces to the case of an interval classifier.

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Some Examples:

- **Rectangles in $\mathbb{R}^2$:** Consider $\mathbf{x} \in \mathbb{R}^2$ and define $\mathcal{C}$ as the class of all rectangles:

$$f(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{x} \in [\alpha_1, \beta_1] \times [\alpha_2, \beta_2] \\ 0 & \text{otherwise} \end{cases}$$

We may assume any three points are not co-linear because if so then it reduces to the case of an interval classifier.

- We claim that $\mathbf{VCD}(\mathcal{C}) = 4$.

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Some Examples:

- **Rectangles in $\mathbb{R}^2$:** Consider $\mathbf{x} \in \mathbb{R}^2$ and define $\mathcal{C}$ as the class of all rectangles:

$$f(\mathbf{x}) = \begin{cases} 1 & \text{if } \mathbf{x} \in [\alpha_1, \beta_1] \times [\alpha_2, \beta_2] \\ 0 & \text{otherwise} \end{cases}$$

We may assume any three points are not co-linear because if so then it reduces to the case of an interval classifier.

- We claim that $\mathbf{VCD}(\mathcal{C}) = 4$.
- What about cubes in $\mathbb{R}^3$?

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

We have seen that the growth function satisfies $\Pi_{\mathcal{C}}(m) = 2^m$ if $m \leq \mathbf{VCD}(\mathcal{C})$. What if $m > \mathbf{VCD}(\mathcal{C})$? We define a function $\Phi_d(m)$ such that $\Phi_d(0) = 1$, $\Phi_0(m) = 1$ for all $d, m \in \mathbb{N}$ and

$$\Phi_d(m) = \Phi_d(m-1) + \Phi_{d-1}(m-1)$$

for all $d, m > 0$. Then we have

$$\Pi_{\mathcal{C}}(m) \leq \Phi_d(m)$$

if $\mathbf{VCD}(\mathcal{C}) = d$. The proof is done using induction on d, m . Also for all m, d ,

$$\Phi_d(m) = \sum_{i=0}^d \binom{m}{i}$$

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Finally, we can bound $\Phi_d(m)$ by the following: When $m \geq d$, we have

Sauer's Lemma:

$$\Pi_C(m) \leq \Phi_d(m) \leq \left(\frac{em}{d}\right)^d.$$

We will show that, the concept of VC dimension will play an important role for Neyman-Pearson classification as we can guarantee some probabilistic bounds on the estimation error.

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Suppose f is the optimal classifier, $f' \in \mathcal{C}$ be any classifier, let

$$(f' \oplus f)(\mathbf{x}) = \begin{cases} 1 & \text{if } f'(\mathbf{x}) \neq f(\mathbf{x}) \\ 0 & \text{otherwise} \end{cases}$$

- Define $\text{err}(f') = \mathbb{P}[(f \oplus f')(\mathbf{x}) = 1]$ as the *misclassification rate*, and $\Delta_f(\mathcal{C}) = \{f' \oplus f, f' \in \mathcal{C}\}$. We claim that $\mathbf{VCD}(\Delta_f(\mathcal{C})) = \mathbf{VCD}(\mathcal{C})$, and let

$$\Delta_{f,\epsilon}(\mathcal{C}) = \{\tilde{f} \in \Delta_f(\mathcal{C}), \mathbb{P}[\tilde{f}(\mathbf{x}) = 1] \geq \epsilon\}$$

i.e, the set of all classifiers in $\mathcal{C}$ with misclassification rate at least ϵ .

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Definition: We say S is an ϵ -net for $\Delta_f(\mathcal{C})$ if $\forall \tilde{f} \in \Delta_{f,\epsilon}(\mathcal{C}), \exists \mathbf{x} \in S$ such that $\tilde{f}(\mathbf{x}) = 1$.

- It means that, if a classifier f' is “bad enough” that has misclassification rate of at least ϵ , then there must exist one point $\mathbf{x}$ such that f' misclassifies the label of $\mathbf{x}$;
- Conversely, if the training set S is an ϵ -net, then any classifier f' that fits the sample well (with zero training error) must have misclassification error $< \epsilon$.

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Now let S be the training set, we will draw two samples from S as follows:

- First draw a sample S_1 of size m , let A be the event that S_1 fails to be an ϵ -net for $\Delta_f(\mathcal{C})$.
- This means $\exists \tilde{f} \in \Delta_{f,\epsilon}(\mathcal{C})$ such that $\tilde{f}(\mathbf{x}) = 0$ for all $\mathbf{x} \in S_1$, but has misclassification rate at least ϵ in the sense of true distribution.
- Fix such a $\tilde{f}$ and draw a second sample S_2 of size m , then we know $\mathbb{P}(\tilde{f}(\mathbf{x}_i) = 1) \geq \epsilon$ for all $\mathbf{x}_i \in S_2$, let

$$X = \sum_{i=1}^m \tilde{f}(\mathbf{x}_i) \sim \text{Binomial}(m, p), p \geq \epsilon$$

then using Chernoff bound we have

$$\mathbb{P}(X < \epsilon m/2) \leq e^{-m\epsilon/8} < \frac{1}{2} \quad \text{for } m = \mathcal{O}(1/\epsilon)$$

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

- Now consider the event B as follows: A sample $S = S_1 \cup S_2$ of size $2m$ with $|S_1| = |S_2| = m$, with the classifier $\tilde{f}$ we fixed such that $|\{\mathbf{x} \in S, \tilde{f}(\mathbf{x}) = 1\}| \geq \epsilon m/2$ and $\tilde{f}(\mathbf{x}) = 0$ for all $\mathbf{x} \in S_1$.
- We have already showed that,

$$\mathbb{P}(B|A) = \mathbb{P}(X \geq \epsilon m/2) \geq \frac{1}{2}$$

hence

$$\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A) \geq \frac{1}{2}\mathbb{P}(A) \quad (\text{Vapnik and Chervonenkis, 1971})$$

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Now it is purely a combinatorial problem to bound $\mathbb{P}(B)$. B is equivalent as saying: If we are given $2m$ balls with $r \geq \epsilon m/2$ are red and the rest are black, then we divide them into 2 sets of m balls each, what is the probability that the first set has no red balls? The answer is simply $\binom{m}{r} / \binom{2m}{r}$, which can be bounded by

$$\frac{\binom{m}{r}}{\binom{2m}{r}} := \prod_{i=0}^{r-1} \frac{m-i}{2m-i} \leq \frac{1}{2^r}$$

Thus by Sauer's lemma we finally have

$$\mathbb{P}(A) \leq 2\mathbb{P}(B) \leq 2 \cdot \left| \Pi_{\Delta_{f,\epsilon}(C)}(S) \right| 2^{-\frac{\epsilon m}{2}} \leq 2 \cdot \left(\frac{2em}{d} \right)^d 2^{-\epsilon m/2}$$

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

From the bound we obtained, for some δ , we solve

$$2 \left(\frac{2em}{d} \right)^d 2^{-\epsilon m/2} \leq \delta$$

and we get

$$\frac{\epsilon m}{2} \geq \frac{d}{\ln 2} \ln \left(\frac{2em}{d} \right) + 1 - \log_2 \delta$$

By choosing m large enough, for instance

$$m \geq \kappa_0 \left(\frac{1}{\epsilon} \log \frac{1}{\delta} + \frac{d}{\epsilon} \log \frac{1}{\epsilon} \right)$$

where κ_0 is some universal constant, we would have $\mathbb{P}(A) \leq \delta$. It means that, if we draw at least m training samples, then with probability at least $1 - \delta$, every consistent classifier f on S will have misclassification error $< \epsilon$.

Vapnik-Chervonenkis Dimension and Probabilistic Bounds

Theorem: Let $\mathcal{C}$ be the class of classifiers with VC dimension $d \geq 2$, then every PAC learning algorithm must use at least $m \geq \frac{d-1}{32\epsilon}$ samples to achieve error less than ϵ with high probability. That is, no learning algorithm can guarantee a classifier with true misclassification error $\leq \epsilon$ unless it uses at least $m \geq \frac{d-1}{32\epsilon}$ training samples.

Relation to Empirical Risks

For a classifier $f \in \mathcal{C}$, we define its empirical misclassification rate as $\hat{R}_n(f)$, and $R(f)$ as the true misclassification rate. It can be shown that

$$\begin{aligned}\mathbb{P}\left(\sup_{f \in \mathcal{C}} |R(f) - \hat{R}_n(f)| > \epsilon\right) &\leq 8\pi_{\mathcal{C}}(2n)e^{-n\epsilon^2/32} \\ &\leq Cn^d e^{-n\epsilon^2/32}\end{aligned}$$

Hence we have

$$\sum_{n=1}^{\infty} \mathbb{P}\left(\sup_{f \in \mathcal{C}} |R(f) - \hat{R}_n(f)| > \epsilon\right) \leq \sum_{n=1}^{\infty} Cn^d e^{-n\epsilon^2/32} < \infty$$

which guarantees the uniform convergence by Borel-Cantelli lemma, hence

$$\sup_{f \in \mathcal{C}} |R(f) - \hat{R}_n(f)| \rightarrow 0 \quad a.s$$

Next, Emile Petit will talk about simulation studies and ROC curve.

- **1.** Computational Learning Theory: VC Dimension, Lecture 3, by Varun Kanade;
- **2.** On the Uniform Convergence of Relative Frequencies of Events to Their Probabilities, by V.N. Vapnik and A. Ya. Chervonenkis;
- **3.** Machine Learning Theory (CSC 482A/581B), Lectures 7 & 8, by Nishant Mahta;